

## Worksheet

1. Find both the parametric and the symmetric form of the line through the point  $(2, -3, 5)$  and in the same direction as  $\langle 4, 1, -7 \rangle$ .
2. Find both the parametric and the symmetric form of the line through the points  $(3, -4, 0)$  and  $(-2, 1, 4)$ .
3. Find both the point normal and standard form of the plane containing the point  $(2, -3, 5)$  and the normal vector  $\langle 5, -1, 4 \rangle$ .
4. Find both the point normal and standard form of the plane containing the points  $(2, -3, 5)$ ,  $(1, 0, 4)$ , and  $(-3, 1, 0)$ .